

Week 2 Notes:

EDA is fundamental to understanding our data :

- Choose appropriate algorithms
- Select relevant features
- Identify data cleaning needs
- Detect outliers or unusual patterns
- Understand the limitations of our potential model

Scatter Plot code:

```
```{r}
Plot premium against client data to test linearity
insurance_model_data %>%
 pivot_longer(cols = -premium) %>%
 ggplot(aes(x=value, y=premium))+
 geom_point()+
 geom_smooth(method="lm")+
 labs(title="Insurance Premium v.s. Client Data")+
 facet_wrap(~name, scales="free")+
 theme_classic()
```
```

This code creates scatterplots that show us whether our assumptions about linear relationships are valid. This is crucial because we're planning to use linear regression. If we see strong non-linear patterns, we might want to approach the data differently !

Breakdown:

- `pivot_longer` restructures our data for plotting
- `ggplot` creates the base plot
- `geom_point()` adds scatter points
- `geom_smooth` adds the blue trend line
- `facet_wrap` creates separate plots for each variable

TODO: please analyze the output value plots and understand what each plot means on your own !

Correlation Code:

```
```{r}
Check correlation between dependent and independent variables
cor(insurance_model_data$premium, subset(insurance_model_data,select=-premium))
```
```

This code:

1. Takes our premium column

2. Correlates it with all other variables
3. `subset(select=-premium)` excludes premium from predictors
4. Returns correlation coefficients between -1 and 1

These numbers quantify what we saw in the plots:

- Smoker: 0.79
- Age: 0.32
- BMI: 0.20
- Children: 0.08
- Region: 0.01

Remember: correlation ranges from -1 to 1, where:

- 1 means perfect positive correlation
- 0 means no linear relationship
- -1 means perfect negative correlation

TODO: interpret the output values and understand what they tell you about the variables

Multilinearity Code:

```
```{r}
Create a correlation matrix to test colinearity
library(corrplot)

Ensure all predictors are numeric
predictors <- insurance_model_data %>%
 select(-premium) %>%
 mutate(
 smoker = as.numeric(smoker),
 region = as.numeric(region),
)

correlation matrix
correlation_matrix <- cor(predictors, use = "pairwise.complete.obs") # Handles NAs
print(correlation_matrix)

corrplot(
 correlation_matrix,
 method = "color",
 tl.col = "black",
 tl.cex = 0.8,
 addCoef.col = "black",
 number.cex = 0.7
)
```
```

Multicollinearity is when predictor variables are highly correlated with each other. This matters in machine learning because:

Model Stability:

- Highly correlated predictors can make our model unstable
- Small changes in data could lead to large changes in coefficients

Feature Importance:

- Multicollinearity makes it difficult to determine the true importance of each feature
- This affects model interpretability

This creates our heatmap showing relationships between all variables. What this code does:

1. `select(-premium)`

- Removes premium column
- We only want relationships between predictors

2. `mutate()`

- Ensures all variables are numeric
- Required for correlation calculation

3. `cor(predictors, use = "pairwise.complete.obs")`

- Creates correlation matrix
- `pairwise.complete.obs` handles any missing values

Why check multicollinearity?

- Highly correlated predictors can destabilize our model
- Makes coefficients harder to interpret
- Helps us decide if we need to remove variables"

TODO: analyze the heatmap and understand why it's important and what it tells us about the multicollinearity of the data

Questions asked:

"Why use scatter plot and not boxplots to understand the data and its patterns in EDA"

Answer: In supervised learning for linear regression, scatterplots were chosen over box plots because they effectively reveal variable relationships and interactions crucial for model development. By visualizing premium against different predictors like age, BMI, and smoking status, the scatterplots show trend lines and variability that highlight how each feature potentially influences the target variable. Unlike boxplots, which summarize distributions, scatterplots provide a more comprehensive view of correlations, helping to understand predictor impacts before building the linear regression model.